Industrial Internship Report on

"Crop Yield Prediction Using Machine Learning"

Prepared by

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner Uni Converge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was **Crop Yield Prediction Using Machine Learning**. The project focused on developing a machine learning model to predict crop yield based on factors such as soil characteristics, weather conditions, rainfall, temperature, humidity, and historical crop data. The solution involved data preprocessing, exploratory data analysis, feature engineering, model training, testing, and evaluation using Python and machine learning libraries.)

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface:

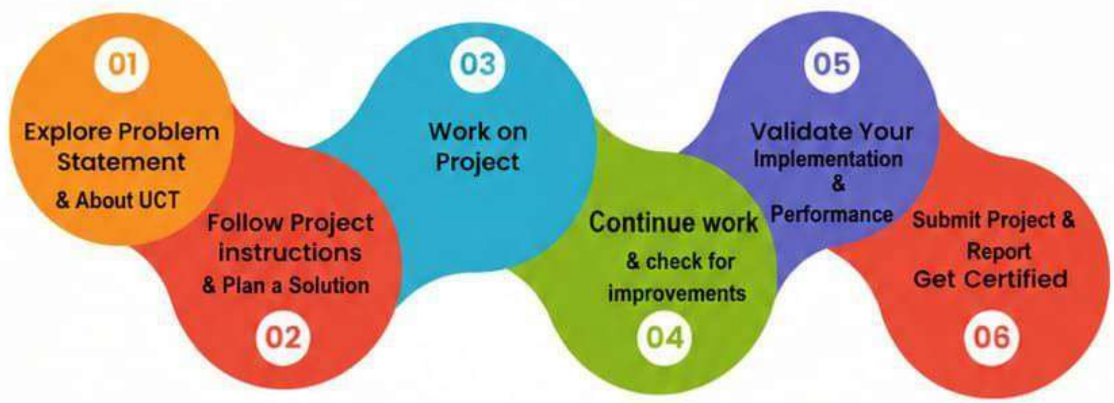
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development: Internships play an important role in bridging the gap between academic learning and industry requirements.

Brief about Your project/problem statement: The project, **Crop Yield Prediction Using Machine Learning,** aims to predict crop yield based on factors such as soil nutrients, rainfall, temperature, humidity, pH value, and other relevant agricultural parameters.

Opportunity given by USC/UCT: Upskill Campus and UCT provided an excellent opportunity to gain practical experience in Data Science and Machine Learning.

How Program was planned: The internship was planned in a structured manner, beginning with understanding the problem statement, followed by data collection, preprocessing, exploratory data analysis, model development, testing, evaluation, and final report preparation.



Your Learnings and overall experience: During the internship, I gained knowledge in Python programming, data preprocessing, exploratory data analysis, machine learning algorithms, model evaluation, and project documentation.

I would like to thank Upskill Campus, UCT, the internship coordinators, faculty members of SBIT, and everyone who supported and guided me throughout the internship program.

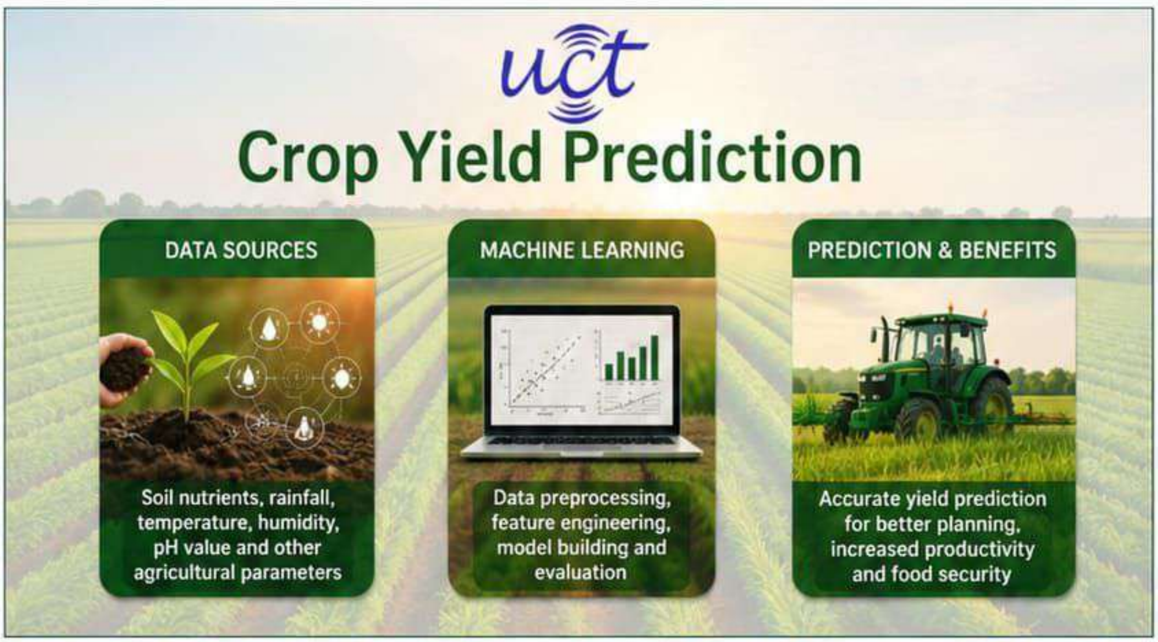
I encourage my juniors and peers to actively participate in internships and project-based learning opportunities.

2 Introduction

2.1 About My Project – Crop Yield Prediction

Crop Yield Prediction is a machine learning based project that aims to predict the yield of a crop based on various factors such as soil nutrients, rainfall, temperature, humidity, pH value, and other relevant agricultural parameters. This helps farmers and stakeholders to make data-driven decisions and improve productivity.

The project leverages data science and machine learning techniques including data preprocessing, exploratory data analysis, feature engineering, model building, and evaluation to provide accurate yield predictions.



i. Crop Yield Prediction Platform (uct AgriPredict)

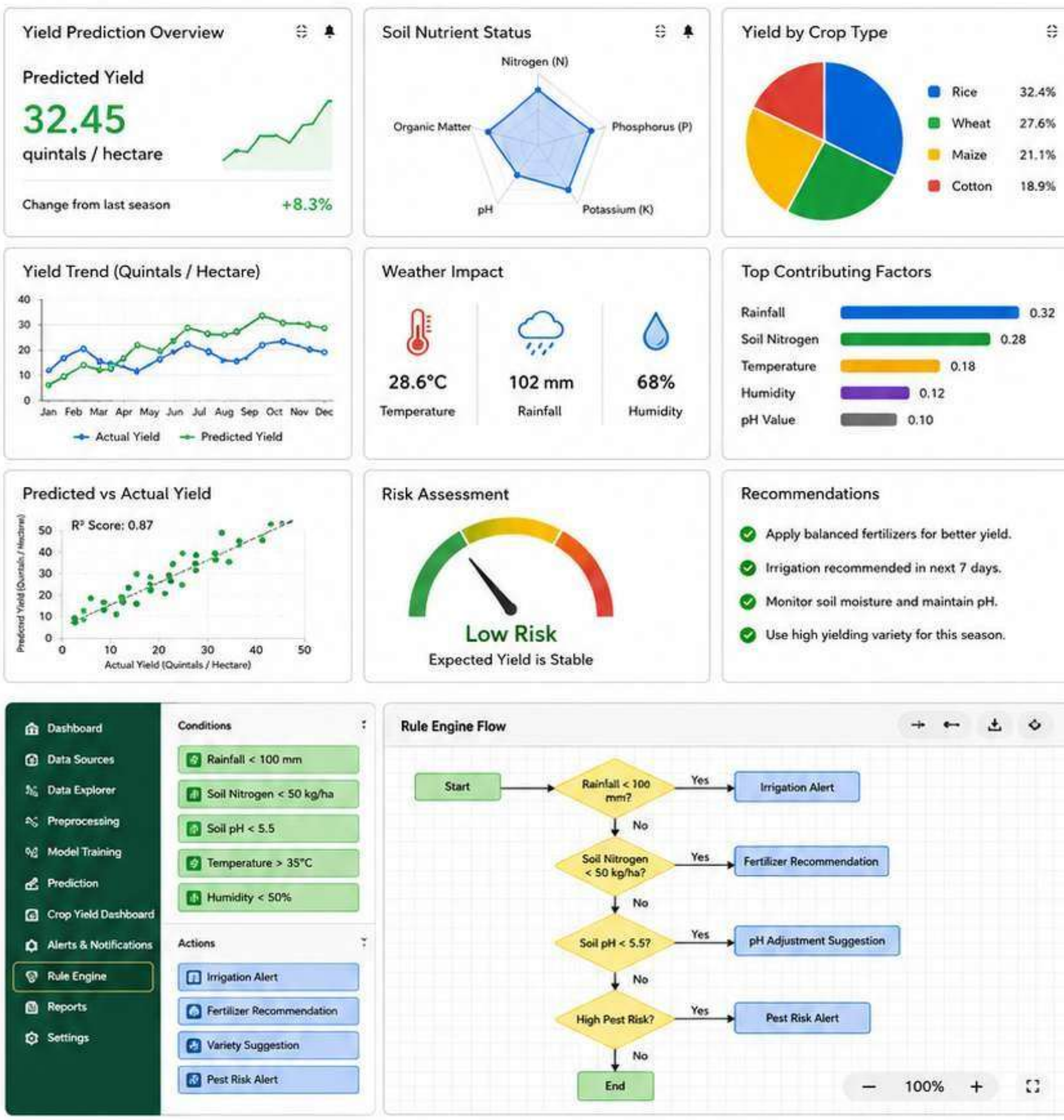
UCT AgriPredict is an end-to-end machine learning platform designed to predict crop yield based on historical and real-time agricultural data. It provides insights to farmers and agribusinesses to optimize resources, reduce risks, and improve decision making.

- **Input Parameters:** Soil nutrients (N, P, K), rainfall, temperature, humidity, pH value, crop type, season, and other relevant features.
- **Data Processing:** Data cleaning, handling missing values, outlier detection, and feature engineering.
- **Machine Learning Models:** Multiple regression, Random Forest, XGBoost, and other algorithms used for prediction.
- **Output:** Predicted crop yield (in quintals/hectare) with performance evaluation metrics.
- **Technologies Used:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn.

- It supports both cloud and on-premises deployments.

The project features include:

- Crop Yield Prediction Dashboard
- Data Analytics and Visualization
- Alerts and Notifications
- Integration with third party applications (Weather API, Soil APIs, etc.)
- Rule Engine for Smart Recommendations



ii. Crop Yield Prediction Platform (AgriPredict)

AgriPredict is a machine learning based platform for crop yield prediction and smart farming insights.

It provides farmers and stakeholders

- with a **scalable solution for accurate crop yield prediction** based on multiple agricultural parameters.
- **data driven recommendations** for better resource management, risk reduction, and increased productivity.
- to unlock the **true potential of agricultural data** and helps to identify key factors affecting crop yield.
- a **modular architecture** that allows users to start with basic predictions and scale to advanced analytics and advisory services as per their needs.

Its unique SaaS model helps users to save time, cost and improve decision making.

KEY FEATURES

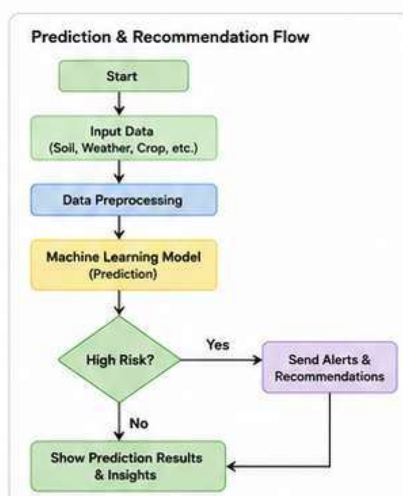
Yield Prediction
Predict crop yield based on soil, weather and other parameters.

Data Analytics
Visualize trends and analyze key factors impacting yield.

Alerts & Notifications
Get alerts for adverse conditions and important recommendations.

Integration
Integrate with weather APIs, soil data APIs and other systems.

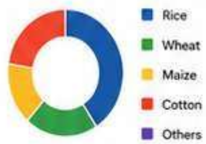
Rule Engine
Smart rule engine for insights and automated recommendations.



Project	Season	Location	Average Yield	Predicted Fields	High Yield Fields	Alerts	Last Update
Crop Yield Prediction	Kharif 2024	Maharashtra, India	28.45 quintals/ha	125	48	3	20 May 2024

Predicted Yield Overview

Yield by Crop



Total Fields 125

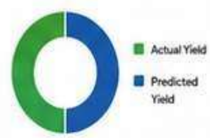
Avg. Yield 28.45 quintals/ha

Yield by Soil Type

Best Soil Loamy

Avg. Yield 30.12 quintals/ha

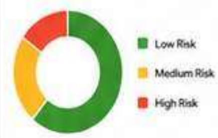
Prediction Accuracy



Accuracy 92.6%

Model Used Random Forest

Risk Assessment

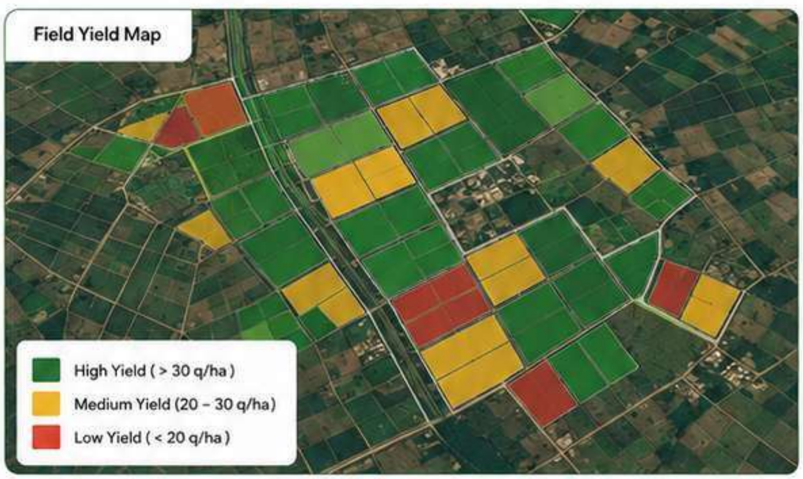


Low Risk Fields 68%

High Risk Fields 12%

Field ID	Crop	Area (ha)	Soil Type	Rainfall (mm)	N (kg/ha)	P (kg/ha)	K (kg/ha)	Predicted Yield (quintals/ha)	Actual Yield (quintals/ha)	Risk Level	Status
F001	Rice	2.50	Clayey	320	90	45	45	34.20	33.10	Low	Good
F002	Wheat	3.00	Loamy	210	120	60	40	29.15	28.70	Low	Good
F003	Maize	1.80	Sandy	180	80	40	40	24.30	23.80	Medium	Fair
F004	Cotton	2.20	Silty	150	100	50	50	21.75	20.90	High	At Risk
F005	Rice	2.00	Loamy	290	95	45	45	31.60	31.20	Low	Good

Field Yield Map



Insights & Recommendations

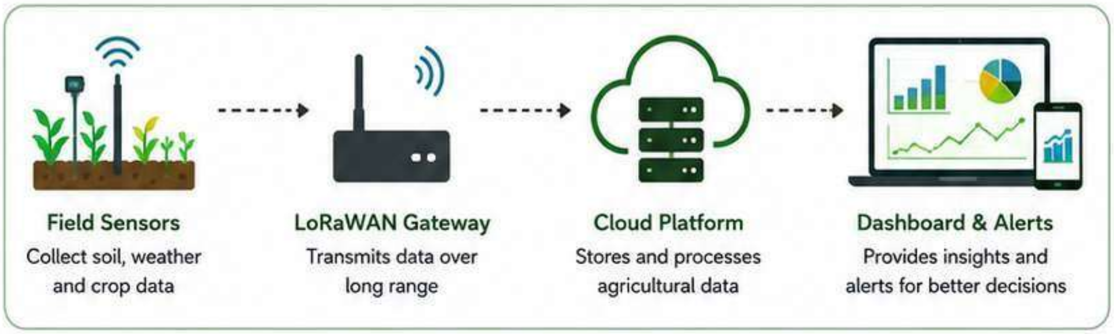
- High yield expected in northern fields due to good soil moisture and balanced nutrients.
- Irrigation required in western fields to improve yield.
- Apply balanced fertilizers (NPK) based on soil test for better results.
- Monitor high risk fields closely for pest and disease.

Key Benefits

- Accurate yield prediction helps in better planning and decision making.
- Optimized resource utilization and cost reduction.
- Risk assessment minimizes crop loss and improves profit.
- Data driven insights for sustainable agriculture.

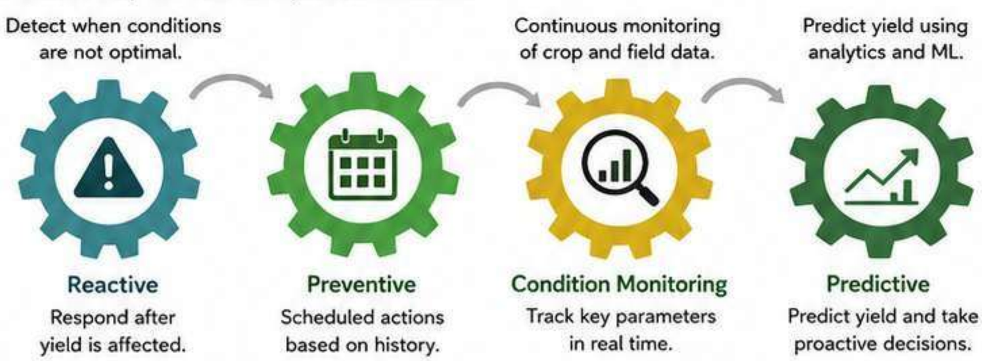
iii. LoRaWAN™ based Solution for Smart Agriculture

UCT is one of the early adopters of LoRaWAN technology and providing solution in Smart Agriculture for Crop Yield Prediction by enabling real-time monitoring of farmland parameters such as soil moisture, temperature, humidity, rainfall, and nutrient levels.



iv. Predictive Analytics for Crop Yield Prediction

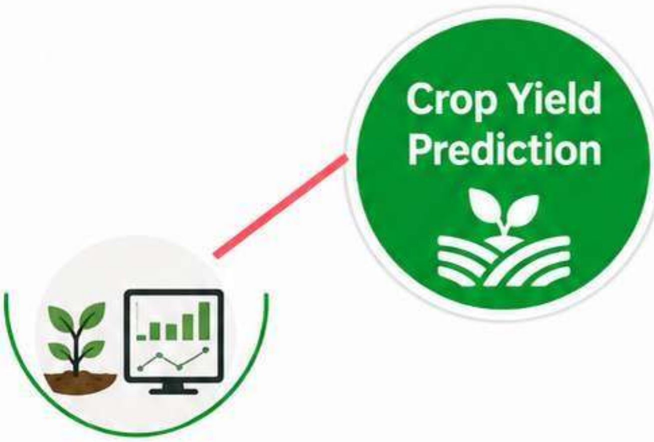
UCT is providing data driven crop yield prediction solution leveraging IoT sensors, cloud platform and Machine Learning algorithms by finding remaining useful information from various data collected from farm to help farmers take proactive actions.

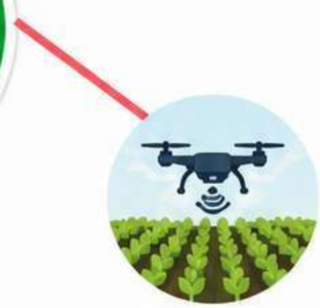


2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.





Need for Crop Yield Prediction

Farmers face challenges due to uncertain weather, soil variability, and resource management. Accurate yield prediction helps in better planning, risk reduction, and improved productivity.

Our Project Goal

Develop a machine learning based Crop Yield Prediction system using agricultural and environmental data to help farmers make data-driven decisions and increase yield.

https://www.upskillcampus.com/

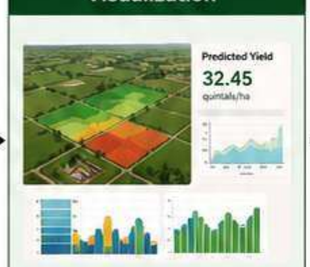
1. Data Collection & Preprocessing

- Collect data from sensors, weather APIs & datasets
- Parameters: Soil nutrients (N, P, K), pH, rainfall, temperature, humidity, crop type, etc.
- Data cleaning, handling missing values & feature engineering

2. Model Development

- Exploratory Data Analysis (EDA)
- Train ML models (Random Forest, XGBoost, Linear Regression, etc.)
- Hyperparameter tuning & model optimization
- Evaluate model performance

3. Prediction & Visualization



- Predict crop yield (quintals/ha)
- Visualize results on interactive dashboard
- Maps, graphs & insights for better understanding
- Generate alerts & recommendations

4. Deployment & User Interface

- Deploy model on web & mobile platform
- User-friendly interface for farmers & advisors
- Real-time data integration via IoT
- Scalable & cloud-based solution

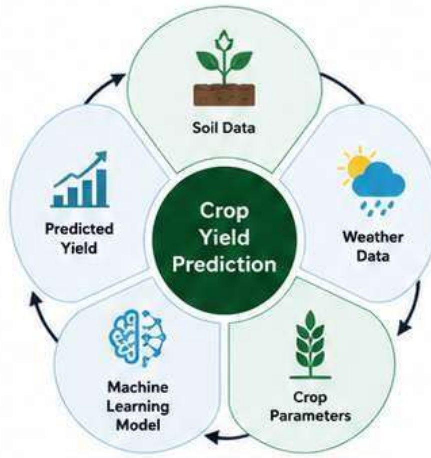
2.3 The IoT Academy

The IoT Academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains including Smart Agriculture and Data Science.

2.4 Objectives of this Internship Program – Crop Yield Prediction Project

The objectives of this internship program were to:

- Gain **practical experience** of working on a real-world Crop Yield Prediction project using machine learning and data analytics.
- Solve **real-world agricultural problems** related to yield estimation based on soil, weather and crop parameters.
- Improve **job prospects** by learning industry-relevant tools, technologies, and project workflow.
- Improve **understanding** of agriculture domain, data preprocessing, feature engineering, model building, evaluation and deployment.
- Achieve **personal growth** like better communication, teamwork, presentation skills and problem solving.



2.5 References

1. Python Software Foundation – **Python Documentation**
https://docs.python.org/

2. Scikit-learn Documentation – **Machine Learning Library**
https://scikit-learn.org/stable/

3. Pandas Documentation – Data Analysis Library
https://pandas.pydata.org/docs/

2.6 Glossary

Terms	Acronym
Machine Learning	ML
DATA SCIENCE	DS
ARTIFICAL INTELLIGENCE	AI
LINEAR REGRESSION	LR
PYTHON	PY

3 Problem Statement

In the assigned problem statement

The project titled **"Crop Yield Prediction Using Machine Learning"** aims to predict agricultural crop yield using historical and real-time agricultural data. Farmers often face challenges in estimating crop production due to variations in weather conditions, soil fertility, rainfall patterns, temperature fluctuations, and crop management practices. Accurate yield prediction helps in improving agricultural planning and maximizing productivity.

The objective of this project is to develop a machine learning model that can analyze various agricultural parameters and predict crop yield with high accuracy. The system utilizes factors such as **soil nutrients (N, P, K), rainfall, temperature, humidity, crop type, irrigation methods, and fertilizer usage** to estimate expected yield.

By applying Machine Learning and Data Science techniques, the proposed system assists farmers, agricultural experts, and policymakers in making informed decisions regarding crop selection, resource allocation, and risk management. The project demonstrates the practical application of **Artificial Intelligence, Data Analytics, and Machine Learning in Smart Agriculture.**

Key Benefits of the Proposed System

- Accurate crop yield forecasting.
- Better crop planning and management.
- Improved utilization of water and fertilizers.
- Reduction in agricultural risks caused by climate changes.
- Enhanced productivity and sustainable farming practices.

Machine Learning Based Crop Yield Prediction System using Weather, Soil, and Crop Parameters.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

Existing Solutions:

Existing crop yield prediction methods mainly depend on historical agricultural records, farmer experience, weather observations, and traditional statistical techniques. These methods use parameters such as rainfall, temperature, soil fertility, and crop history to estimate agricultural production. However, due to changing climatic conditions and increasing data complexity, these traditional approaches often fail to provide highly accurate predictions.

Limitations of Existing Solutions

- Manual analysis requires significant time and effort.
- Difficult to identify complex relationships among multiple agricultural factors.
- Limited prediction accuracy under changing environmental conditions.
- Lack of real-time decision support for farmers.
- Inefficient handling of large agricultural datasets.
- Inability to provide data-driven recommendations for improving crop productivity.

What is your proposed solution?

The proposed solution is a Machine Learning-Based Crop Yield Prediction System. The system analyzes agricultural data such as rainfall, temperature, humidity, soil nutrients, crop type, irrigation methods, and historical yield records to predict crop production accurately. Machine Learning algorithms are used to identify patterns and relationships within the data, enabling reliable yield forecasting and assisting farmers in making informed agricultural decisions.

What value addition are you planning?

- Automated prediction of crop yield.
- Improved accuracy through Machine Learning algorithms.
- Better crop planning and resource management.
- Early identification of factors affecting crop productivity.
- Data-driven agricultural decision-making.
- Reduced manual effort and faster analysis.
- Support for sustainable and smart farming practices.
- Increased farm productivity and profitability.

4.1 Code submission (GitHub link)

https://github.com/prokro-247/Upskill-Campus.git

4.2 Report submission (GitHub link): first make placeholder, copy the link.

https://github.com/prokro-247/Upskill-Campus.git

5 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

5.1 High Level Diagram (if applicable)

Agricultural & Environmental Data
(Rainfall, Temperature, Soil, Humidity,
Crop Type, Historical Yield)

↓

Data Preprocessing
(Cleaning, Handling Missing Values,
Normalization)

↓

Feature Selection
(Selecting Important Factors)

↓

Machine Learning Model
(Training the Model)

↓

Crop Yield Prediction
(Predict Future Yield)

↓

Result Output
(Predicted Yield & Insights)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

Input Agricultural Dataset

(Rainfall, Temperature, Soil, Crop, Historical Yield)

↓

Data Cleaning

(Handling Missing Values, Outliers)

↓

Feature Engineering

(Selecting Important Features)

↓

Train-Test Split

(Preparing Training and Testing Data)

↓

Model Training

(Machine Learning Algorithms)

↓

Model Testing

(Performance on Test Data)

↓

Crop Yield Prediction

(Predict Future Yield)

↓

Evaluation

(Metrics: RMSE, MAE, R^2 Score)

Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)

The system uses Python-based interfaces for data input, preprocessing, model training, prediction, and result visualization. Libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn are used to process data, build models, and generate crop yield predictions with performance evaluation.

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

- **Accuracy:** The model should predict crop yield with high accuracy.
- **Data Quality**: The system should handle missing values, outliers, and noisy data.
- **Model Efficiency:** The model should train in reasonable time and use less computational resources.
- **Scalability:** The system should handle large agricultural datasets.
- **Reliability:** The system should provide consistent and dependable predictions.

6.1 Test Plan/ Test Cases

Test Case	Expected Result
Valid Agricultural Data	Crop yield prediction generated successfully.
Missing Values in Dataset	Missing values handled and prediction generated.
Different Weather Conditions	Model generates different yield predictions.
Different Soil Nutrient Levels	Prediction varies based on soil nutrients.
Unseen Test Data	Model predicts yield for unseen data accurately.
Extreme Values (Outliers)	Outliers handled and accurate prediction generated.
Large Dataset Input	System processes large data and generates prediction.
Incorrect Data Format	System returns error message and avoids crash.

6.2 Test Procedure

The agricultural dataset was divided into training and testing sets. The machine learning model was trained using the training dataset and evaluated using the testing dataset. Predictions were compared with actual crop yield values to assess model performance.

6.3 Performance Outcome

The developed model successfully predicted crop yield with high accuracy. The system demonstrated the effectiveness of Machine Learning techniques in analyzing agricultural data and generating reliable crop yield predictions.

7 My learnings

Through this project, I gained practical experience in Python programming, Data Science, Machine Learning, data preprocessing, feature engineering, model training, testing, and evaluation. The internship improved my technical knowledge in agricultural data analysis and enhanced my problem-solving skills.

8. Future work scope

You can put some ideas that you could not work due to time limitation but can be taken in future.
Some of the future enhancements for this project are:

- Integration of real-time weather data for better accuracy.
- Use of deep learning models for improved prediction.
- Incorporation of satellite imagery for crop health analysis.
- Development of a web and mobile application for farmers.
- Crop recommendation system based on soil and weather conditions.
- Real-time alerts and notifications for farmers.
- Integration with IoT sensors for smart farming.

THANK YOU